Pitching Performance and Elbow Loading

Exploratory analysis of Driveline's OpenBiomechanics Project · independent project, not a peer-reviewed publication.

411
recorded pitches

100
athletes

2–5
pitches per athlete

> **Research question.** Do completed-pitch rotational summaries add predictive information beyond age, body mass, and height for pitch speed and peak internal elbow varus moment in held-out athletes?

Design and data trust

One-to-one source pitch join; 100 sessions. Missing focal values: 0; 25 other POI columns contain missing observations. Missing outcomes remain missing.

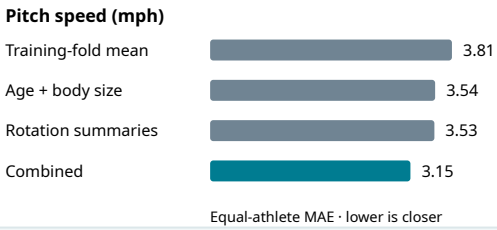
5 shared athlete-grouped folds (seed 42); four fixed baselines; ridge alpha 10. Training-only median imputation, missingness indicators, and standardization. Predictors are age/mass/height, three rotational peaks, or all six. Outcomes, IDs, and filenames are excluded.

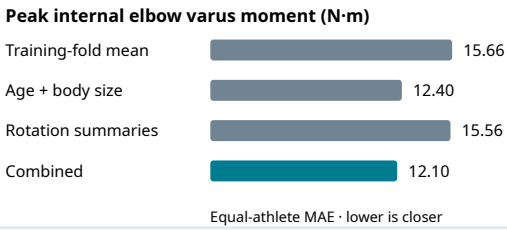
The comparison plan followed viewed Phase 1 results. This is exploratory, not preregistered or independently confirmatory. Predictors describe a completed pitch, not a pre-pitch forecast.

Saved athlete-held-out results

Equal-athlete MAE: average absolute error within athlete, then average athletes equally. Lower is closer.

Model	Pitch speed (mph)	Peak internal elbow varus moment (N·m)
Training-fold mean	3.807	15.662
Age + body size	3.542	12.404
Rotation summaries	3.530	15.557
Combined	3.153	12.104





Pitch speed

Combined − age/body-size MAE: **-0.389 mph**.

Conditional 95% interval: [-0.749, -0.029] mph.

Within-source error contrast; no causal intervention or external-performance claim.

Elbow loading

Combined − age/body-size MAE: **-0.299 N·m**.

Conditional 95% interval: [-1.218, +0.628] N·m.

Increment remains uncertain: interval includes zero.

Intervals use 5,000 paired athlete resamples of fixed out-of-fold errors and exclude full model-refitting uncertainty. The 10 overlapping refitted partitions are descriptive sensitivity checks, not independent replications or confidence intervals.

> The source cohort is primarily collegiate and each athlete has one released assessment. These results do not establish external performance, longitudinal development, causal coaching effects, or injury risk.

Source code · Research dashboard · Full exploratory results

Source: Driveline Baseball Research & Development, OpenBiomechanics Project. Source data, definitions, and this source-derived brief retain CC BY-NC-SA 4.0 plus the additional professional-sports / financial-analysis affiliation exclusion. Those uses require the licensor's specific written paid license. Original software is MIT; that license does not grant restricted data or model-use rights. Full source terms.

Evidence, reproducibility, and next research question

What this work establishes

Source provenance / join	**Passed source audit** Pinned files; one-to-one pitch join; recorded missingness.
Released scalar reproduction	**12/12 source checks passed** Finite maxima of supplied traces compared with supplied POIs; not new ground truth.
Within-source prediction	**Athlete-held-out assessment** 5 shared athlete-grouped folds; learned preprocessing stays in training folds.
Software behavior	**See release verification** Automated and browser checks exercise integrity, missingness, grouping, and frozen evaluation.
Independent inverse dynamics	**Not performed** Joint moments are released source outputs; no independent dynamics reconstruction.
External cohort validation	**Not performed** Local freeze/evaluate tooling is implemented; no independent cohort has been supplied.
Marker–signal synchronization	**Unverified** A matching sample grid does not establish a shared time origin or camera alignment.
Injury / causal intervention	**Outside scope** No injury outcomes, tissue-specific forces, or causal mechanics intervention.

Released-signal reproduction audit

12 of 12 source comparisons passed. Full-trace signed finite maxima versus published POIs; absolute tolerance 0.0001 in each quantity's native units reflects source rounding, not physical accuracy or independent biomechanical validation. Trace gaps remain missing.

Pitch	Released quantity	Trace max	POI	Unit	Check
1393_3	Hip–shoulder separation	15.9429	15.9429	deg	passed
1393_3	Canonical shoulder z rotation velocity	4320.9953	4320.9953	deg/s	passed
1393_3	Torso axial rotation velocity	968.5634	968.5634	deg/s	passed
1393_3	Internal elbow varus moment	91.9207	91.9207	N·m	passed
1623_3	Hip–shoulder separation	21.0338	21.0338	deg	passed
1623_3	Canonical shoulder z rotation velocity	4100.6570	4100.6570	deg/s	passed
1623_3	Torso axial rotation velocity	1006.1412	1006.1412	deg/s	passed
1623_3	Internal elbow varus moment	96.9136	96.9136	N·m	passed
2916_4	Hip–shoulder separation	36.1736	36.1736	deg	passed
2916_4	Canonical shoulder z rotation velocity	5192.6458	5192.6458	deg/s	passed
2916_4	Torso axial rotation velocity	1322.9304	1322.9304	deg/s	passed
2916_4	Internal elbow varus moment	130.0088	130.0088	N·m	passed

A defensible next study

Evaluate frozen definitions and models in an independently authorized cohort with compatible measurement conventions. Establish athlete independence with the data owner and freeze development choices before viewing evaluation outcomes. Report cohort shift and unfavorable results. Local audit/freeze/evaluate tooling is implemented; this study remains unperformed.

Contribution and assistance

Project framing and requirements by Owen Garrett; code and documentation developed with AI assistance. Personal review, execution, and modifications should be described accurately in the contribution record. This brief makes no claim of sole implementation. Contribution record.

Inspect the evidence

Pinned OBP source · Measurement audit · Reproduction instructions · Release checks

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OBP commit: 44b98dae05cceb016f080ab39d105c85b8639084
Analysis SHA-256: 83c751ae903ce5f62a12e0678159c275e456c004f88c1f164caa191cf63b0b14
Producing project revision: 77dea84e6d03985fb145172f1f721e7168909480
```

Net elbow varus moment is joint-level loading, not isolated UCL force. Peak hip–shoulder separation is not specifically separation at foot plant. Marker and processed-signal clocks remain unverified.

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